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Combination washing and drying laundry appliance having a filter cleaning system for removing lint during a laundry cycle

Abstract

A washing and drying appliance includes a tub having an air outlet. A drum rotationally operates within the tub. An air handling system draws process air from inside the tub and through a lint filter disposed within the air outlet that filters particulate from the process air. During a filter cleaning phase, a fluid delivery system delivers impulse amounts of wash fluid toward a sump of the tub. The impulse amounts of wash fluid are maintained between an outer surface of the drum and an inside surface of the tub. The drum operates to direct the wash fluid around the drum and along the inside surface of the tub and over an interior surface of the lint filter to remove particulate from the lint filter. A fluid pump is coupled with the sump to direct wash fluid and particulate to an external outlet.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

(1) The present disclosure generally relates to laundry appliances, and more specifically, to a combination washing and drying appliance having a filter cleaning system that can be used to remove particulate from a lint filter while articles to be dried are contained within a rotating drum. Combination washing and drying appliances include both washing and drying capabilities such that a fluid delivery system and an airflow path are each included within the appliance for providing these washing and drying functions within a single appliance.

SUMMARY OF THE DISCLOSURE

(2) According to one aspect of the present disclosure, a combination washing and drying appliance includes a tub having a wall and positioned within an outer cabinet. The tub includes an air outlet through an upper portion of the wall. A drum rotationally operates within the tub. An air handling system is attached to the wall of the tub and that draws process air from inside the tub and through the air outlet to an airflow path. A lint filter is disposed within the air outlet and that filters particulate from the process air entering the airflow path. A fluid delivery system includes a valve that operates to deliver wash fluid into the inside of the tub. A sump is defined within a lower portion of the tub and below the drum. The sump includes a drain that moves wash fluid from the inside of the tub to an external outlet. During a filter cleaning phase of the appliance, the fluid delivery system delivers impulse amounts of the wash fluid to deliver the wash fluid to the sump and to maintain the wash fluid between an outer surface of the drum and an inside surface of the tub. The drum operates to direct the wash fluid in the sump around the drum and along the inside surface of the tub and over an interior surface of the lint filter to remove particulate from the lint filter. A fluid pump coupled with the sump directs the wash fluid and the particulate to the external outlet.

(3) According to another aspect of the present disclosure, a method for operating a combination washing and drying appliance includes completing a first drying cycle with articles present within a rotating drum. A second drying cycle is selected immediately after the first drying cycle. The articles are distributed along an inner surface of the rotating drum by rotating the rotating drum at a first speed. Wash fluid is added to a tub that surrounds the rotating drum by adding impulse amounts of wash fluid that travel along an inside surface of the tub by surface adhesion. The wash fluid is circulated as circulated fluid along the inside surface of the tub using air ventilation produced by the rotating drum rotating at a second speed that is faster than the first speed. Particulate is removed from a lint filter using the circulated fluid. The circulated fluid and the particulate are collected within the sump by decelerating the rotating drum to a third speed to allow wash fluid and the particulate to fall under force of gravity into the sump of the tub, the third speed being slower than the second speed. The wash fluid and the particulate are drained from the sump by activating a fluid pump attached to the sump. The second drying cycle is activated.

(4) According to yet another aspect of the present disclosure, a method for cleaning a lint filter located within a tub of a combination washing and drying appliance during performance of a drying phase of a laundry cycle where articles are located within a perforated drum that rotates within the tub includes steps of sensing a blockage in the lint filter, initiating an interruption in the drying phase, distributing the articles along an inner surface of the perforated drum by rotating the perforated drum at a first speed, adding wash fluid to the tub, circulating the wash fluid as circulated fluid along an inside surface of the tub by rotating the perforated drum at a second speed that is faster than the first speed, washing the lint filter using the circulated fluid to remove particulate from a surface of the lint filter, collecting the circulated fluid and the particulate within a lower portion of the tub by decelerating the perforated drum to allow the circulated fluid and the particulate to fall according to force of gravity and draining a sump to deliver the wash fluid and particulate to an external drain.

(5) These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

- (2) FIG. 1 is a front elevational view of a combination washing and drying appliance incorporating an aspect of the filter cleaning assembly;
- (3) FIG. 2 is a perspective view of a combination washing and drying appliance with the outer cabinet removed;
- (4) FIG. 3 is a perspective view of an aspect of a combination washing and drying appliance with the outer cabinet removed;
- (5) FIG. 4 is a cross-sectional view of the combination washing and drying appliance of FIG. 3 taken along line IV-IV;
- (6) FIG. 5 is a top perspective view of an aspect of the combination washing and drying appliance of FIG. 3 with the blower removed and showing an aspect of the air outlet with the lint filter installed therein;
- (7) FIG. 6 is a partially exploded perspective view of the combination washing and drying appliance of FIG. 5 and showing a lint filter removed from the air outlet;
- (8) FIG. 7 is a schematic perspective view of an aspect of the filter cleaning assembly and showing a distributing step for balancing the articles contained within the drum along the inner surface of the drum;
- (9) FIG. 8 is a schematic view of the combination washing and drying appliance of FIG. 7 and showing the drum rotating at a high rate of speed to circulate wash fluid through an interstitial space defined between the drum and the tub;
- (10) FIG. 9 is a schematic cross-sectional view of the tub and drum of FIG. 8 and showing a collection phase of the filter cleaning assembly where wash fluid and particulate are collected within a sump without allowing the wash fluid and particulate to enter the inside of the drum;
- (11) FIG. 10 is a schematic cross-sectional view of the drum and tub of FIG. 9 and showing a draining phase where the wash fluid and particulate are delivered away from the tub via the fluid pump;
- (12) FIG. 11 is a schematic diagram showing operation of various components of the combination washing and drying appliance during performance of the filter cleaning phase of a particular laundry cycle;
- (13) FIG. 12 is a schematic diagram illustrating a process for determining whether the filter cleaning phase should be initiated and also showing performance of the filter cleaning phase;
- (14) FIG. 13 is a linear flow diagram illustrating a method for operating a combination washing and drying appliance; and
- (15) FIG. 14 is a linear flow diagram illustrating a method for cleaning a lint filter located within a tub of a combination washing and drying appliance during performance of a drying phase of a laundry cycle and where articles are located within a perforated drum that rotates within the tub.
- (16) The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles described herein.

DETAILED DESCRIPTION

(17) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to a combination washing and drying appliance having a lint filter installed within a tub and configured to perform a filter cleaning phase that removes particulate from the lint filter during performance of a drying cycle and without removing articles from the rotating drum and also without allowing fluid to enter into the drum that may saturate the articles being dried. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

(18) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,”

“vertical,” “horizontal,” and derivatives thereof shall relate to the disclosure as oriented in FIG. 1. Unless stated otherwise, the term “front” shall refer to the surface of the element closer to an intended viewer, and the term “rear” shall refer to the surface of the element further from the intended viewer. However, it is to be understood that the disclosure may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(19) The terms “including,” “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises a . . .” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

(20) Referring to FIGS. 1-10, reference numeral 10 generally refers to a laundry appliance, and typically a combination washing and drying appliance that can perform both washing and drying functions with respect to the articles 12 being processed. In this manner, a user can place articles 12 within a drum 14 of the appliance 10 and the appliance 10 will perform both washing and drying functions such that two appliances 10 are not needed to process the articles 12.

(21) According to the various aspects of the device, as exemplified in FIGS. 1-10, the appliance 10 includes a tub 20 having a wall 22, where the tub 20 is positioned within an outer cabinet 24. The tub 20 includes an air outlet 26 that is positioned through an upper portion 28 of the wall 22. The drum 14 rotationally operates within the tub 20 and typically about a horizontal or angled rotational axis 30. An air handling system 32 that includes a blower 34 for moving process air 36 through an airflow path 38 is attached to the wall 22 of the tub 20. A lint filter 40 is disposed within the air outlet 26. The lint filter 40 filters particulate 42 from the process air 36 that enters the airflow path 38 from the tub 20. Accordingly, during operation of the blower 34 for the air handling system 32, process air 36 is drawn from within the tub 20 and moves through the air outlet 26 to be filtered by the lint filter 40. During this process, particulate 42 is captured within the lint filter 40 and can accumulate on an interior surface 44 of the lint filter 40 that faces the inside of the tub 20. A fluid delivery system 46 is also attached to the tub 20. The fluid delivery system 46 includes a valve 48 that operates to deliver wash fluid 50, typically water, into the inside of the tub 20. A sump 52 is defined within the lower portion 54 of the tub 20 and below the drum 14. The sump 52 includes a drain 56 that moves wash fluid 50 and particulate 42 from inside of the tub 20 to an external outlet 58. This external outlet 58 can be in the form of an external drain 56, a removable water bottle, combinations thereof and other similar mechanisms for disposing, recycling or recirculating the wash fluid 50.

(22) Referring again to FIGS. 1-10, the appliance 10 described herein includes the lint filter 40 within the air outlet 26 and operates in the absence of a condenser attached to the tub 20. In certain alternative appliances, this condenser forms a part of the air path and assists in removing lint particles. This condenser is typically positioned along the exterior surface of the tub. The appliance 10 described herein is manufactured to operate without this condenser and utilizes the lint filter 40 in the air outlet 26 for separating particulate 42 from the process air 36.

(23) Referring again to FIGS. 1-10, the combination washing and drying appliance 10 is configured to perform a filter cleaning phase 70. During the filter cleaning phase 70, the fluid delivery system 46 delivers impulse amounts 72 of a wash fluid 50 into the tub 20. These impulse amounts 72 of wash fluid 50 are typically very small amounts that slowly move along the inside surface of the tub 20 using the phenomena of surface adhesion. Using surface adhesion, this wash fluid 50 moves along the inside surface without being splashed or dropping onto the drum 14 or through the

perforated wall **76** that forms the drum **14**. Accordingly, during the impulse operation of the valve **48**, the wash fluid **50** entering into the tub **20** is maintained separate from the drum **14** so as to not saturate the articles **12** contained within the drum **14**. Accordingly, during the filter cleaning phase **70**, the articles **12** remain within the drum **14** and the wash fluid **50** is maintained within an interstitial space **78** inside of the tub **20** and outside of the drum **14**.

(24) During this filter cleaning phase **70**, the drum **14** operates to direct the wash fluid **50** within the drum **14** and around the outside surface **90** of the tub **20**. To accomplish this, the drum **14** rotates at a high rate of speed using the process of air ventilation **162** within the interstitial space **78** defined between the drum **14** and the tub **20**. As the drum **14** rotates at a high rate of speed, the wash fluid **50** defines circulated fluid **92** that is circulated along the inside surface of the tub **20** and also circulated over the interior surface **44** of lint filter **40**. Accordingly, the circulated fluid **92** is moved across the interior surface **44** of the lint filter **40** to remove particulate **42** from the lint filter **40**. The wash fluid **50** and particulate **42** are then collected within the sump **52**. A fluid pump **94** that is coupled with the sump **52** directs the wash fluid **50** and particulate **42** to the external outlet **58**. Initially, the circulated fluid **92** includes the wash fluid **50**. As the filter cleaning phase **70** is performed, the wash fluid **50** moves across the interior surface **44** of the lint filter **40** and removes the particulate **42** therefrom. In this manner, the circulated fluid **92**, over time, includes greater and greater amounts of the particulate **42** that is added to the wash fluid **50**.

(25) During operation of the appliance **10**, typically a drying phase **100**, the air handling system **32** operates to move process air **36** through the airflow path **38**. In this manner, the air handling system **32** operates to move process air **36** from inside the tub **20** and through the air outlet **26** and into the airflow path **38** of the laundry appliance **10**. As the process air **36** moves through the air outlet **26**, the process air **36** and particulate **42** move through the lint filter **40**. At the lint filter **40**, the particulate **42** is separated from the process air **36** and is captured and temporarily retained on the interior surface **44** of the lint filter **40**. In certain aspects of the device, the air handling system **32** can also include a heater as well as other air conditioning systems that can be used for heating, cooling and dehumidifying the process air **36** within the appliance **10**. Accordingly, as the process air **36** moves through the air outlet **26** and the air handling system **32**, the process air **36** is filtered and conditioned for delivery back into the tub **20** via an air inlet **102** at the front **104** of the tub **20**.

(26) Referring again to FIGS. **3-10**, the filter cleaning phase **70** is configured to operate while the articles **12** being dried are contained within a processing space **110** defined within the perforated wall **76** of the drum **14**. Using the rotational operation of the drum **14** as well as the impulse operation of the valve **48** for delivering small amounts of wash fluid **50** into the tub **20**, the wash fluid **50** is maintained within the interstitial space **78** and separated from the drum **14**. In this manner, the wash fluid **50** is prevented from entering into the drum **14** and saturating the articles **12**.

(27) It is contemplated that certain amounts of wash fluid **50** may inadvertently move into the inside of the drum **14** through the perforated wall **76** via splashing and other inadvertent movements of the wash fluid **50**. However, using the rotation of the drum **14** and the impulse operation of the valve **48**, this amount of fluid is kept to a minimum such that substantially all of the wash fluid **50** is maintained between the rotating drum **14** and the tub **20**. The term “substantially” in this regard is meant to reflect that approximately five percent or less than approximately five percent of the wash fluid **50** used during the filter cleaning phase **70** enters into the drum **14**. As will be described more fully below, the drum **14** is typically configured to rotate during the entire filter cleaning phase **70**. Accordingly, wash fluid **50** and particulate **42** that engages the outer surface of the rotating drum **14** is typically shed away from the rotating drum **14** through centrifugal force **120**.

(28) Referring again to FIGS. **3-10**, it is contemplated that a filter cleaning phase **70** is initiated between sequential drying phases **100**. By way of example, and not limitation, where a first drying cycle **130** is performed, the articles **12** contained within the drum **14** may not be completely dried

or may not be dried to the level desired by the user of the appliance **10**. Accordingly, a second drying cycle **132** can be operated immediately after the first drying cycle **130**. After completion of the first drying cycle **130**, the lint filter **40** typically has at least some particulate **42** captured within the interior surface **44** thereof. Accordingly, use of the filter cleaning phase **70** is utilized for removing this accumulated particulate **42** from the lint filter **40** to ensure proper movement of the process air **36** through the drum **14**, the tub **20** and the airflow path **38** for drying the articles **12**. After completion of the filter cleaning phase **70**, where particulate **42** is removed from the lint filter **40**, the selected second drying cycle **132** is performed. It is contemplated that upon selection of the second drying cycle **132** to be performed immediately after the first drying cycle **130**, the filter cleaning phase **70** is automatically operated as an initial portion of the second drying cycle **132**.

(29) Referring again to FIGS. **3-10**, it is also contemplated that the filter cleaning phase **70** can be performed during an interruption **140** of the drying cycle **130**. The interruption **140** can be initiated by a sensor that detects an amount of particulate **42** that is entrapped on the interior surface **44** of the lint filter **40**. The filter cleaning phase **70** can be performed during this interruption **140**. After the filter cleaning phase **70** is complete, the drying cycle **130** being performed is continued until completed, or until such time as the filter cleaning phase **70** is performed again.

(30) Referring again to FIGS. **3-10**, where the filter cleaning phase **70** is performed during interruption **140** in a drying phase **100** being performed, articles **12** are present within the processing space **110** during this interruption **140**. As described herein, during performance of the filter cleaning phase **70**, wash fluid **50** used during this filter cleaning phase **70** is maintained within the interstitial space **78** defined between the drum **14** and the tub **20**. The use of impulse operations of the valve **48** and continuous rotation of the drum **14**, or substantially continuous rotation of the drum **14**, during this filter cleaning phase **70** helps to maintain the wash fluid **50** and the particulate **42** within this interstitial space **78**. Accordingly, the articles **12** being dried are not saturated or substantially saturated with the wash fluid **50** used during the filter cleaning phase **70**.

(31) Referring now to FIGS. **5-12**, the filter cleaning phase **70** includes a low speed balancing phase **150** wherein the drum **14** rotates at a first speed **156**. This low speed balancing phase **150** is used to distribute the articles **12** contained within the drum **14** along an inner surface **152** of the drum **14**. During this low speed balancing phase **150**, the articles **12** are distributed around the inner surface **152** to define a balanced state **154** of the articles **12**. In this balanced state **154** of the articles **12**, the drum **14** is configured to smoothly rotate about the rotational axis **30** of the drum **14** without significant wobbling or oscillation. Wobbling or oscillation of the drum **14** and the tub **20** may result in unwanted splashing of the wash fluid **50** onto the rotating drum **14** during performance of the filter cleaning phase **70**. Accordingly, by placing the articles **12** in the balanced state **154**, the relative motion of the drum **14** and the tub **20** within the cabinet **24** remains substantially limited throughout the filter cleaning phase **70** so as to not disrupt the surface adhesion of the wash fluid **50** against the inside surface of the tub **20**. This allows the impulse amounts **72** of wash fluid **50** to move along the inside surface of the tub using surface adhesion.

(32) Referring again to FIGS. **7-12**, during the low speed balancing phase **150**, the valve **48** operates intermittently, and through the impulse operations, to deliver the impulse amounts **72** of wash fluid **50**. As described herein, these impulse amounts **72** of wash fluid **50** travel along with the inside surface of the tub **20** and toward the sump **52** using the phenomena of surface adhesion. Typically, a fluid inlet of the fluid delivery system **46** is located below a midpoint of the tub **20** defined by the rotational axis **30** to allow the fluid to run along the inside surface of the tub **20**. Using surface adhesion, it is contemplated that the fluid inlet can be located at a position above the midpoint of the tub **20**. Surface adhesion allows the wash fluid **50** to flow along and adhere to the inside surface of the tub **20**.

(33) According to the various aspects of the device, the amount of wash fluid **50** utilized during the filter cleaning phase **70** can vary. By way of example, and not limitation, the valve **48** can operate one time per ten seconds to release small impulse amounts **72** of wash fluid **50** into the tub **20**.

Over the course of approximately 10 minutes, approximately 5 deciliters (approximately 0.5 liters) of wash fluid **50** can be delivered into the tub **20**. It is contemplated that greater or lesser amounts of wash fluid **50** can be delivered into the tub **20** for accomplishing the filter cleaning phase **70**. Typically the amount of wash fluid **50** delivered into the tub **20** will depend at least upon the amount of space available within the tub **20** and below the drum **14**. Additionally, to maintain the balanced state **154** of the articles **12** contained within the drum **14**, the drum **14** is configured to rotate continuously as the valve **48** operates the impulse activations to slowly deliver the wash fluid **50** into the tub **20**. This rotation of the drum **14** to achieve the balanced state **154** of the articles **12** can be a rotational speed of approximately 70 rotations per minute to approximately 150 rotations per minute. In certain aspects of the device, approximately 90 rotations per minute are utilized to achieve the balanced state **154** of the articles **12**.

(34) Referring again to FIGS. 7-12, the filter cleaning phase **70** also includes a high speed cleaning phase **160** that is characterized by the drum **14** rotating at a second speed **164** that is a high rate of speed and faster than the first rate of speed. At this second speed **164**, the movement of the drum **14** relative to the interstitial space **78** causes a process of air ventilation **162** that moves the wash fluid **50** as circulated fluid **92** through the interstitial space **78** and along the inside surface of the tub **20**. As this circulated fluid **92** is moved along the inside surface of the tub **20**, this circulated fluid **92** is also moved along the interior surface **44** of the lint filter **40** for separating particulate **42** from the lint filter **40**.

(35) According to various aspects of the device, this high speed cleaning phase **160** of the drum **14** is configured to operate at approximately 800 revolutions per minute. Other rotational speeds of the drum **14** are also contemplated. These rotational speeds are typically sufficient to generate the process of air ventilation **162** within the interstitial space **78** that causes the circulation of the wash fluid **50** along the inside surface of the tub **20**. To achieve the proper movement of the circulated fluid **92** within the interstitial space **78** and along the inside surface of the tub **20**, this high speed cleaning phase **160** can last from approximately 60 seconds to approximately 100 seconds. It is also contemplated that other time periods for operating the high speed cleaning phase **160** can be utilized.

(36) Referring again to FIGS. 7-12, after completion of the high speed cleaning phase **160**, the drum **14** operated a collecting phase **170** where the drum **14** slows to a third speed **172** that is slower than the second speed **164** utilized during the high speed cleaning phase **160**. The third speed **172** may be the same as the first speed **156** used during the low speed balancing phase **150** or may be between the first speed **156** and the second speed **164**. During this collecting phase **170** of the filter cleaning phase **70**, the drum **14** slows such that the effect of air ventilation **162** is diminished. Accordingly, the wash fluid **50** that has been circulated along the inside surface of the tub **20** falls from the inside surface of the tub **20** or moves along the inside surface of the drum **14** under the force of gravity. Certain portions of this wash fluid **50** as well as the particulate **42** may fall onto an outside surface **90** of the drum **14**. Because the drum **14** is rotating at the third speed **172**, this wash fluid **50** and particulate **42** are shed from the outside surface **90** of the drum **14** utilizing centrifugal force **120**. Accordingly, the wash fluid **50** that falls upon the outer surface of the drum **14** is quickly shed from the outside surface **90** and directed to the inside surface of the tub **20** to be directed to the sump **52**.

(37) As described herein, the wash fluid **50** is shed from the outside surface **90** of the drum **14** and moved toward the inside surface of the tub **20**. This wash fluid **50** can move along the tub **20** using the phenomena of surface adhesion. During this collecting phase **170** where the drum **14** rotates at the third speed **172**, the fluid pump **94** that is attached to the sump **52** for the tub **20** begins to operate. Accordingly, as the wash fluid **50** and particulate **42** are collected within the sump **52**, the fluid pump **94** extracts the wash fluid **50** and particulate **42** and moves it to the external outlet **58**. By operating the fluid pump **94** during the performance of the collecting phase **170** of the filter cleaning phase **70**, as wash fluid **50** and particulate **42** are moved into the sump **52**, these materials

are expediently moved to the external outlet **58** such that large amounts of wash fluid **50** and particulate **42** cannot collect within the sump **52**. Large amounts of wash fluid **50** and particulate **42** may result in splashing of wash fluid **50** onto the drum **14**. By keeping only minimal amounts of wash fluid **50** within the sump **52**, the chances of this splashing occurring are kept to a minimum. (38) Referring now to the schematic diagrams of FIGS. **11-12** that show exemplary aspects of the operation of the appliance **10** and how the filter cleaning phase **70** can be integrated into the laundry cycles. FIG. **11** reflects an exemplary operation of the appliance **10** during the filter cleaning phase **70**. Additionally, FIG. **12** reflects a flow diagram for determining whether the filter cleaning phase **70** should be operated or not.

(39) According to the flow diagram of FIG. **12**, step **402** includes performing a first cycle that includes a drying phase **100**, such as the first drying cycle **130**. This may include a washing and drying phase **100** or just a drying phase **100**. Step **404** includes selecting a second cycle. The decision of step **406** determines whether this second cycle is only a wash cycle, is a wash and dry cycle or is only a dry cycle. Only where the second cycle is the second drying cycle **132** will the filter cleaning phase **70** be initiated. Where the cycle includes only washing or both washing and drying, the filter will be cleaned during the washing cycle and the filter cleaning phase **70** will not be initiated. According to step **408**, the filter cleaning phase **70** is initiated and the clothes are desired to be dried beyond the level provided by the first drying cycle **130**. Step **410** reflects that the second drying cycle **132** is performed. The decision of step **412** includes the use of a sensor for determining the amount of particulate **42** captured by the lint filter **40**. Where the lint filter **40** is blocked or blocked a certain percentage, the decision of step **412** can reinstate the filter cleaning phase **70** to remove accumulated amounts of particulate **42** from the lint filter **40**. When the filter cleaning phase **70** is complete, step **410** for performing the second drying cycle **132** is completed and the cycle is ended (step **414**).

(40) Referring again to FIGS. **11-12**, the filter cleaning phase **70** includes step **502** of draining any wash fluid **50** from the sump **52** that may be contained within the sump **52** or the remainder of the tub **20**. Step **504** includes using impulse activations of the valve **48** to provide small impulse amounts **72** of wash fluid **50** into the tub **20** and using surface adhesion for directing this fluid towards the sump **52**. An unbalanced check is performed at step **506** for determining whether the articles **12** are properly distributed into the balanced state **154**. During the unbalanced check, also referred to herein as the low speed balancing phase **150**, the drum **14** is rotated at the first speed **156** to distribute the articles **12** evenly throughout the inner surface **152** of the drum **14**. Step **508** includes continuing to fill wash fluid **50** into the tub using the impulse activations of the valve **48**. This step **508** can occur for various periods of time. By way of example, and not limitation, this activation of the valve **48** can continue for approximately 30 seconds as the drum **14** continues to rotate at the first speed **156** indicative of the low speed balancing phase **150**. When the proper amount of wash fluid **50** is contained within the drum **14**, step **510** includes spinning the drum **14** at the second speed **164** to perform the high speed cleaning phase **160**.

(41) Referring again to FIGS. **11-12**, toward the end of this high speed cleaning phase **160**, the fluid pump **94** can be activated to begin to remove wash fluid **50** and captured particulate **42** from the tub **20** and to the external outlet **58**. Step **512** includes operating the fluid pump **94** for draining the wash fluid **50** and captured particulate **42** from the sump **52** and moving this wash fluid **50** and captured particulate **42** to the external outlet **58**. As the fluid pump **94** operates, the drum **14** can be slowed to the third speed **172**. To ensure that wash fluid **50** does not enter into the interior of the drum **14**, the drum **14** can be rotated at a fourth speed to utilize the centrifugal force **120** for shedding wash fluid **50** from the outside surface **90** of the drum **14**. By way of example, and not limitation, this fourth speed can include approximately RPM, and typically approximately 27 RPM. This can occur for approximately 90 seconds.

(42) According to the various aspects of the device, the various speeds that are utilized for rotating the drum **14** and time periods which the drum **14** is rotated can vary depending on certain factors.

Such factors can include, but are not limited to, the size of the tub **20** and drum **14** for the appliance **10**, the amount of articles **12** being processed, the type of articles **12** being processed and other similar factors that may affect the speed at which the drum **14** has been rotated and the amount of time in which the drum **14** has rotated to deliver the amount of wash fluid **50** desired. The time taken to deliver the amount of wash fluid **50** needed for the filter cleaning phase **70** can also vary. (43) Referring now to FIGS. **1-13**, having described various aspects of the device, a method **600** is disclosed for operating a combination washing and drying appliance **10**, in particular, operating a filter cleaning phase **70** of a combination washing and drying appliance **10**. According to the method **600**, step **602** includes completing a first drying cycle **130** where articles **12** are present within the drum **14** that rotates within the tub **20**. A second drying cycle **132** is selected (step **604**). This second drying cycle **132** is selected to be performed immediately after the first drying cycle **130**, without an intervening washing cycle performed between. After selecting the second drying cycle **132**, step **606** includes distributing the articles **12** along an inner surface **152** of the rotating drum **14** by rotating the rotating drum **14** at the first speed **156**. As the articles **12** are distributed to define the balanced state **154**, as described herein (or after the balanced state is achieved), wash fluid **50** is added to the tub **20** by adding impulse amounts **72** of wash fluid **50** that travel along an inside surface **74** of the tub **20** using surface adhesion (step **608**). The wash fluid **50** is then circulated in the form of circulated fluid **92** along the inside surface **74** of the tub **20** using air ventilation **162** produced by the rotation of the drum **14** rotating at a second speed **164** that is faster than the first speed **156** (step **610**). Particulate **42** is then removed from the lint filter **40** using the circulated fluid **92** (step **612**). The circulated fluid **92** and the particulate **42** removed from the lint filter **40** are collected within the sump **52**. This collection can occur as the circulated fluid **92** and the particulate **42** move around the inside surface **74** of the tub **20**. This collection can also occur by decelerating the drum **14** to the third speed **172** (step **614**). By decelerating the drum **14**, this allows the wash fluid **50** and the particulate **42** to fall under the force of gravity toward the sump **52** of the tub **20**. As described herein, the third speed **172** is slower than the second speed **164**. The wash fluid **50** and the particulate **42** are drained from the sump **52** (step **616**). This drain **56** occurs through the use of a fluid pump **94** that is attached to the sump **52**. After the wash fluid and particulate **42** are drained from the sump **52**, the second drying cycle **132** is activated (step **618**). As described herein, the step of adding wash fluid **50** occurs at least during the step of distributing the articles **12** along the inner surface **152** of the surface of the drum **14**. This step of adding wash fluid **50** can occur during the entire step of distributing the articles **12**, or can occur during only a portion of this distribution step.

(44) Referring now to FIGS. **1-12** and **14**, having described various aspects of the device for cleaning a lint filter **40**, a method **700** is disclosed for cleaning a lint filter **40** located within a tub **20** of a combination washing and drying appliance **10**. This method **700** is operated during the performance of a drying phase **100** of a laundry cycle where articles **12** are located within a perforated drum **14** that rotates within the tub **20**. This method **700** includes a step of sensing a blockage within the lint filter **40** (step **702**). This sensing can occur through various sensors that can include, but are not limited to, airflow sensors, voltage sensors coupled with the motor for the blower **34**, temperature sensors, optical sensors, and other similar sensors that can be utilized for monitoring an amount of particulate **42** or other blockage that is entrapped on the interior surface **44** of the lint filter **40**. According to the method **700**, step **704** includes initiating an interruption **140** to the drying phase **100** so that the filter cleaning phase **70** can be performed. A step **706** includes distributing the articles **12** along the inner surface **152** of the perforated drum **14** by rotating the perforated drum **14** at the first speed **156**. Once the articles **12** are distributed into the balanced state **154**, wash fluid **50** is added to the tub **20** (step **708**). As described herein, this step of adding wash fluid **50** occurs by operating a valve **48** using impulse operations for providing very small amounts of wash fluid **50** that can be moved toward the sump **52** under surface adhesion. According to the method **700**, step **710** includes circulating the wash fluid **50** as circulated fluid **92**

along the inside surface **74** of the tub **20** by rotating the drum **14** at the second speed **164** that is faster than the first speed **156**. As described herein, the movement of the drum **14** at the second speed **164** utilizes the effective air ventilation **162** for moving the circulated fluid **92** along the inside surface **74** of the tub **20** and over the interior surface **44** of the lint filter **40** for separating particulate **42** from the lint filter **40** (step **712**). Once the lint filter **40** is washed using the circulated fluid **92**, the circulated fluid **92** and the particulate **42** are collected within a lower portion **54** of the tub **20** (step **714**). This lower portion **54** of the tub **20** is typically characterized by the sump **52**. Additionally, the drum **14** is decelerated to minimize the effect of air ventilation **162** such that the wash fluid **50** and particulate **42** can fall under the force of gravity toward the sump **52**. As the wash fluid **50** and particulate **42** collect within this sump **52**, step **716** includes draining the sump **52** to deliver the wash fluid **50** and particulate **42** to the external drain **56**.

(45) According to various aspects of the device, the adding step (step **708**) typically occurs during at least a portion of the distributing step (step **706**). Additionally, the draining step (step **716**) typically occurs during at least a portion of the collecting step (step **714**).

(46) According to the various aspects of the device, this filter cleaning phase **70** can be operated during consecutive drying cycles. Where a washing cycle is utilized between two drying cycles, the washing cycle will necessarily result in fluid being distributed throughout the tub **20** and removing particulate **42** from the lint filter **40**. In this instance where two drying phases **100** occur consecutively, no washing phase occurs such that particulate **42** collects on the lint filter **40**. Additionally, because two drying phases **100** are being conducted consecutively, it is typical that the same load of articles **12** are being processed within an appliance **10**. Accordingly, using the filter cleaning phase **70** described herein, the articles **12** being dried within the drum **14** do not need to be removed during performance of the filter cleaning phase **70**. Additionally, because of the configuration of the rotating drum **14** and the speed at which the drum **14** rotates as well as the impulse activations of the valve **48** for distributing wash fluid **50**, the wash fluid **50** is contained within the interstitial space **78** defined between the drum **14** and the tub **20**. Accordingly, little if any wash fluid **50** is able to enter into the drum **14** for partially saturating the articles **12** contained therein. Accordingly, the drying level achieved by the first drying cycle **130** is not frustrated by the application of additional wash fluid **50** being added into the drum **14**. Use of the filter cleaning phase **70** prevents infiltration of wash fluid **50** into the drum **14** such that little to no re-drying needs to occur.

(47) According to various aspects of the device, use of the filter cleaning phase **70** is typically conducted within a combination washing and drying appliance **10**. These appliances **10** can include a recirculating airflow path **38** that recirculates the same process air **36** within the cabinet **24** for the appliance **10**. This filter cleaning phase **70** can also be used in combination washing and drying appliances **10** that utilize a vented system where process air **36** is vented to areas external of the appliance **10**.

(48) According to another aspect of the present disclosure, a combination washing and drying appliance includes a tub having a wall and positioned within an outer cabinet. The tub includes an air outlet through an upper portion of the wall. A drum rotationally operates within the tub. An air handling system is attached to the wall of the tub and that draws process air from inside the tub and through the air outlet to an airflow path. A lint filter is disposed within the air outlet and that filters particulate from the process air entering the airflow path. A fluid delivery system includes a valve that operates to deliver wash fluid into the inside of the tub. A sump is defined within a lower portion of the tub and below the drum. The sump includes a drain that moves wash fluid from the inside of the tub to an external outlet. During a filter cleaning phase of the appliance, the fluid delivery system delivers impulse amounts of the wash fluid to deliver the wash fluid to the sump and to maintain the wash fluid between an outer surface of the drum and an inside surface of the tub. The drum operates to direct the wash fluid in the sump around the drum and along the inside surface of the tub and over an interior surface of the lint filter to remove particulate from the lint

filter. A fluid pump coupled with the sump directs the wash fluid and the particulate to the external outlet.

(49) According to another aspect, the filter cleaning phase is configured to operate while articles being dried are within a processing space defined within a perforated wall of the drum.

(50) According to yet another aspect, rotational operation of the drum during the filter cleaning phase maintains substantially all of the wash fluid outside of the drum and within an interstitial space defined between the outer surface of the drum and the inside surface of the tub.

(51) According to another aspect of the present disclosure, the filter cleaning phase is initiated after a first drying cycle and before a second drying cycle. The second drying cycle is directly after the first drying cycle.

(52) According to another aspect, the filter cleaning phase is performed during an interruption in a drying cycle. The interruption is initiated by a sensor that detects an amount of the particulate on the inside surface of the lint filter and the drying cycle is continued after the filter cleaning phase is completed.

(53) According to yet another aspect, articles are present within the drum during the interruption and during the filter cleaning phase that is initiated during the interruption.

(54) According to another aspect of the present disclosure, the filter cleaning phase includes a low speed balancing phase that distributes articles within the drum along an inner surface of the drum to define a balanced state of the articles.

(55) According to another aspect of the present disclosure, during the low speed balancing phase, the valve operates intermittently to deliver impulse amounts of the wash fluid that travel through surface adhesion along the inside surface of the tub and to the sump.

(56) According to another aspect, the filter cleaning phase includes a high speed cleaning phase characterized by the drum rotating at a high rate of speed that uses air ventilation within the interstitial space to direct the wash fluid along the inside surface of the tub and over the lint filter.

(57) According to yet another aspect, the filter cleaning phase includes a draining phase characterized by the drum decelerating and using centrifugal force to direct the wash fluid toward the inside surface of the tub and further characterized by activation of the fluid pump to deliver the wash fluid and the particulate to the external outlet.

(58) According to another aspect of the present disclosure, the drum includes a perforated wall that allows wash fluid from moving between a processing space within the drum and the interstitial space between the drum and the tub. During operation of the filter cleaning phase, the wash fluid is maintained within the interstitial space and outside of the rotating drum.

(59) According to another aspect, a method for operating a combination washing and drying appliance includes completing a first drying cycle with articles present within a rotating drum. A second drying cycle is selected immediately after the first drying cycle. The articles are distributed along an inner surface of the rotating drum by rotating the rotating drum at a first speed. Wash fluid is added to a tub that surrounds the rotating drum by adding impulse amounts of wash fluid that travel along an inside surface of the tub by surface adhesion. The wash fluid is circulated as circulated fluid along the inside surface of the tub using air ventilation produced by the rotating drum rotating at a second speed that is faster than the first speed. Particulate is removed from a lint filter using the circulated fluid. The circulated fluid and the particulate are collected within the sump by decelerating the rotating drum to a third speed to allow wash fluid and the particulate to fall under force of gravity into the sump of the tub, the third speed being slower than the second speed. The wash fluid and the particulate are drained from the sump by activating a fluid pump attached to the sump. The second drying cycle is activated.

(60) According to yet another aspect, the step of adding wash fluid happens during the step of distribution of the articles along the inside surface of the tub.

(61) According to another aspect of the present disclosure, the step of decelerating the rotating drum results in a portion of the circulated fluid falling onto an outer surface of the rotating drum.

Rotations of the rotating drum at the third speed results in the circulated fluid being shed from the outer surface of the rotating drum via centrifugal force.

(62) According to another aspect, the circulated fluid is maintained outside of the rotating drum to prevent the circulated fluid from saturating the articles contained within the rotating drum.

(63) According to yet another aspect, a method for cleaning a lint filter located within a tub of a combination washing and drying appliance during performance of a drying phase of a laundry cycle where articles are located within a perforated drum that rotates within the tub includes steps of sensing a blockage in the lint filter, initiating an interruption in the drying phase, distributing the articles along an inner surface of the perforated drum by rotating the perforated drum at a first speed, adding wash fluid to the tub, circulating the wash fluid as circulated fluid along an inside surface of the tub by rotating the perforated drum at a second speed that is faster than the first speed, washing the lint filter using the circulated fluid to remove particulate from a surface of the lint filter, collecting the circulated fluid and the particulate within a lower portion of the tub by decelerating the perforated drum to allow the circulated fluid and the particulate to fall according to force of gravity and draining a sump to deliver the wash fluid and particulate to an external drain.

(64) According to another aspect of the present disclosure, the step of adding the wash fluid to the tub is accomplished using impulse activations of a fluid valve to deliver impulse amounts of the wash fluid along the inside surface of the tub using surface adhesion.

(65) According to another aspect, the step of collecting the circulated fluid and the particulate includes rotating the perforated drum at a speed that sheds a portion of the circulated fluid and the particulate toward the inside surface of the tub using centrifugal force.

(66) According to yet another aspect, the adding step occurs during the distributing step.

(67) According to another aspect of the present disclosure, the draining step occurs during the collecting step.

(68) It will be understood by one having ordinary skill in the art that construction of the described disclosure and other components is not limited to any specific material. Other exemplary embodiments of the disclosure disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.

(69) For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.

(70) It is also important to note that the construction and arrangement of the elements of the disclosure as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other

substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

(71) It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present disclosure. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

Claims

1. A combination washing and drying appliance comprising: a tub having a wall and positioned within an outer cabinet, the tub having an air outlet defined within an upper portion of the wall; a drum that rotationally operates within the tub; an air handling system attached to the wall of the tub and that draws process air from inside the tub and through the air outlet of the wall of the tub and to an airflow path; a lint filter that is disposed within the air outlet and that filters particulate from the process air entering the airflow path; a fluid delivery system having a valve attached to the tub and that operates to deliver wash fluid into the inside of the tub; and a sump defined within a lower portion of the tub and below the drum, the sump having a drain that moves wash fluid from the inside of the tub to an external outlet, wherein: during a filter cleaning phase of the appliance, the valve of the fluid delivery system delivers impulse amounts of the wash fluid to deliver the wash fluid to the sump by traveling along an inside surface of the tub by surface adhesion to maintain the wash fluid between an outer surface of the drum and the inside surface of the tub; the drum operates to direct the wash fluid in the sump around the drum and along the inside surface of the tub and over an interior surface of the lint filter to remove particulate from the lint filter; a fluid pump coupled with the sump directs the wash fluid and the particulate to the external outlet.
2. The combination washing and drying appliance of claim 1, wherein the filter cleaning phase is configured to operate while articles being dried are within a processing space defined within a perforated wall of the drum.
3. The combination washing and drying appliance of claim 1, wherein rotational operation of the drum during the filter cleaning phase maintains substantially all of the wash fluid outside of the drum and within an interstitial space defined between the outer surface of the drum and the inside surface of the tub.
4. The combination washing and drying appliance of claim 1, wherein the filter cleaning phase is initiated after a first drying cycle and before a second drying cycle, wherein the second drying cycle is directly after the first drying cycle.
5. The combination washing and drying appliance of claim 1, wherein the filter cleaning phase is performed during an interruption in a drying cycle, wherein the interruption is initiated by a sensor that detects an amount of the particulate on the inside surface of the lint filter and wherein the drying cycle is continued after the filter cleaning phase is completed.
6. The combination washing and drying appliance of claim 5, wherein articles are present within the drum during the interruption and during the filter cleaning phase that is initiated during the interruption.
7. The combination washing and drying appliance of claim 1, wherein the filter cleaning phase includes a low speed balancing phase that distributes articles within the drum along an inner surface of the drum to define a balanced state of the articles.
8. The combination washing and drying appliance of claim 7, wherein during the low speed balancing phase, the valve operates intermittently to deliver the impulse amounts of the wash fluid that travel through surface adhesion along the inside surface of the tub and to the sump, wherein the valve is positioned within a lower position of the wall of the tub relative to the air outlet.
9. The combination washing and drying appliance of claim 3, wherein the filter cleaning phase

includes a high speed cleaning phase characterized by the drum rotating at a high rate of speed that uses air ventilation within the interstitial space to direct the wash fluid from the sump and along the inside surface of the tub and over the lint filter.

10. The combination washing and drying appliance of claim 1, wherein the filter cleaning phase includes a draining phase characterized by the drum decelerating and using centrifugal force to direct the wash fluid toward the inside surface of the tub and further characterized by activation of the fluid pump to deliver the wash fluid and the particulate to the external outlet.

11. The combination washing and drying appliance of claim 1, wherein the drum includes a perforated wall that allows wash fluid from moving between a processing space within the drum and the interstitial space between the drum and the tub, and wherein during operation of the filter cleaning phase, the wash fluid is maintained within the interstitial space and outside of the rotating drum.

12. A method for operating a combination washing and drying appliance, the method comprising steps of: completing a first drying cycle with articles present within a rotating drum; selecting a second drying cycle immediately after the first drying cycle; distributing the articles along an inner surface of the rotating drum by rotating the rotating drum at a first speed; adding wash fluid to a tub that surrounds the rotating drum by adding impulse amounts of wash fluid that travel along an inside surface of the tub by surface adhesion to a sump; circulating the wash fluid from the sump as circulated fluid along the inside surface of the tub using air ventilation produced by the rotating drum rotating at a second speed that is faster than the first speed; removing particulate from a lint filter using the circulated fluid; collecting the circulated fluid and the particulate within the sump by decelerating the rotating drum to a third speed to allow wash fluid and the particulate to fall under force of gravity into the sump of the tub, the third speed being slower than the second speed; draining the wash fluid and the particulate from the sump by activating a fluid pump attached to the sump; and activating the second drying cycle.

13. The method of claim 12, wherein the step of adding wash fluid happens during the step of distribution of the articles along the inside surface of the tub.

14. The method of claim 12, wherein the step of decelerating the rotating drum results in a portion of the circulated fluid falling onto an outer surface of the rotating drum, and wherein rotations of the rotating drum at the third speed results in the circulated fluid being shed from the outer surface of the rotating drum via centrifugal force.

15. The method of claim 12, wherein the circulated fluid is maintained outside of the rotating drum to prevent the circulated fluid from saturating the articles contained within the rotating drum.

16. A method for cleaning a lint filter located within a tub of a combination washing and drying appliance during performance of a drying phase of a laundry cycle where articles are located within a perforated drum that rotates within the tub, the method comprising steps of: sensing a blockage in the lint filter; initiating an interruption in the drying phase; distributing the articles along an inner surface of the perforated drum by rotating the perforated drum at a first speed; adding wash fluid to the tub for delivery to a sump within a lower portion of the tub; circulating the wash fluid from the sump as circulated fluid along an inside surface of a wall of the tub by rotating the perforated drum at a second speed that is faster than the first speed; washing the lint filter using the circulated fluid to remove particulate from a surface of the lint filter; collecting the circulated fluid and the particulate within the sump of the tub by decelerating the perforated drum to allow the circulated fluid and the particulate to fall according to force of gravity; and draining the sump to deliver the wash fluid and particulate to an external drain.

17. The method of claim 16, wherein the step of adding the wash fluid to the tub is accomplished using impulse activations of a fluid valve disposed within the wall of the tub to deliver impulse amounts of the wash fluid along the inside surface of the wall of the tub using surface adhesion.

18. The method of claim 16, wherein the step of collecting the circulated fluid and the particulate includes rotating the perforated drum at a speed that sheds a portion of the circulated fluid and the

particulate toward the inside surface of the tub using centrifugal force.

19. The method of claim 16, wherein the adding step occurs during the distributing step.

20. The method of claim 16, wherein the draining step occurs during the collecting step.
